41/50

CY1010_Part 1 & 2 Total Marks: 50x1 Date: 5th September 2026

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1. Which combination of Earth systems is described as interacting to maintain conditions suitable for life?
A. Atmosphere, hydrosphere, lithosphere and biosphere
B. Stratosphere, mesosphere, ionosphere and exosphere
C. Lithosphere, magnetosphere, thermosphere and exosphere
D. Hydrosphere, thermosphere, ionosphere and magnetosphere

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2. Without the atmosphere, Earth's average temperature would be approximately:
A. +15 °C
B. −18 °C
C. −63 °C
D. +4 °C

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3. Why does the atmosphere play an important role in maintaining Earth's temperature?
A. It completely blocks visible radiation
B. It produces oxygen through combustion
C. It contributes to the greenhouse effect
D. It prevents solar radiation from reaching Earth

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4. Which phenomenon occurs when light changes direction without a change in its energy/frequency?
A. Absorption
B. Rayleigh scattering
C. Emission
D. Ionization

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5. Which region of the electromagnetic spectrum is strongly absorbed by greenhouse gases according to the lecture?
A. Infrared
B. Visible
C. Radio
D. Gamma rays

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6. An atmospheric window is best described as:
A. A region where sunlight is completely absorbed
B. A portion of the electromagnetic spectrum transmitted through the atmosphere
C. A region containing only visible radiation
D. A region where greenhouse gases are absent

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7. Which activity is identified as a significant source of methane emissions?
A. Rice cultivation
B. Glass recycling
C. Solar-cell production
D. Nuclear fission

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8. Which gas has a global warming potential (GWP) of approximately 22,800 times that of CO_2?
A. CH_4
B. N_2O
C. HFC
D. SF_6

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9. Why are carbon particles from diesel engines important in atmospheric chemistry?
A. They remain in the atmosphere for thousands of years
B. They can absorb heat and reflect incoming light
C. They destroy all atmospheric oxygen
D. They are the main source of methane

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10. Tropospheric ozone is formed primarily through reactions involving:
A. O_2 and N_2 only
B. NOx and VOCs in sunlight
C. CO_2 and H_2O in sunlight
D. None of the above

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11. The general combustion equation for a hydrocarbon C_xH_y produces:
A. CO and H_2 only
B. C and H_2O only
C. CO_2 and H_2O
D. CH_4 and O_2

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12. CNG is obtained by compressing natural gas to approximately:
A. 0.2–0.5 MPa
B. 2–5 MPa
C. 20–25 MPa
D. 200–250 MPa

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13. Steam methane reforming (SMR) converts methane and steam primarily into:
A. CO and H_2
B. CO_2 and O_2
C. CH_4 and O_2
D. C and H_2O

$CH_4 + H_2O \rightarrow CO + 3H_2$

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14. In a hydrogen fuel cell, the overall reaction is:
A. $H_2 \rightarrow 2H^+ + 2e^-$
B. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ + electrical energy
C. $H_2 + O_2 \rightarrow H_2O_2$ + electrical energy
D. $H_2O \rightarrow H_2 + O_2$

$H_2 \rightarrow 2H^+ + 2e^-$
$2e^- + \frac{1}{2}O_2 \rightarrow O^{2-}$

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15. The reaction used to convert triglycerides into fatty acid methyl esters is called:
A. Hydrolysis
B. Hydrogenation
C. Transesterification
D. Depolymerization

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16. Which pore size corresponds to a microporous material according to the IUPAC classification used in the lecture?
A. < 2 nm
B. 2–50 nm
C. > 50 nm
D. 50–100 nm

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17. Which catalyst is identified in the lecture for the ring-opening copolymerization of epoxides with CO_2 to form polycarbonates?
A. $FeCl_3$
B. $ZnEt_2$
C. $NiBr_2$
D. None of the above

18. Which sequence correctly describes the basic operation of a photocatalyst?
A. Light absorption → electron–hole generation → charge separation/transfer → surface redox reaction
B. Light absorption → heat generation → catalyst decomposition → oxidation
C. Charge recombination → light emission → adsorption → reduction
D. Adsorption → excitation → charge generation → desorption

19. Which property of MOFs is particularly useful for selective gas adsorption?
A. Their inability to modify pore structure
B. Their tunable pore size and chemical environment
C. Their complete absence of internal pores
D. Their high electrical conductivity alone

20. Why can increasing temperature reduce the surface coverage in physisorption?
A. Physisorption is generally exothermic and weakly bound
B. Higher temperature creates additional adsorption sites
C. Molecules become chemically bonded to the surface
D. None of the above

21. Which renewable-energy technology utilizes the temperature difference between warm surface seawater and colder deep seawater?
A. Tidal energy
B. Ocean Thermal Energy Conversion
C. Wind energy
D. None of the above

22. Which of the following best represents the role of a catalyst in a chemical reaction?
A. It increases the equilibrium constant by changing ΔG°
B. It lowers the activation energy and increases the reaction rate without being consumed overall
C. It permanently changes the products of a thermodynamically fixed reaction
D. All the three conditions mentioned in A, B and C is correct

23. Which CO_2-utilization process involves the reaction of phenol-derived sodium phenoxide with CO_2?
A. Urea synthesis
B. Polycarbonate synthesis
C. Salicylic acid synthesis
D. None of the above

24. Which option represents the most direct approach to waste minimization?
A. Recycling waste after disposal
B. Incinerating all waste
C. Source reduction to prevent waste generation
D. All the above criteria

25. Earth's atmosphere contains Carbon dioxide ~ 0.04% (volume in %). 80% (approx.) of mass of Earth's atmosphere is in the troposphere. [The radius of Earth ~ 6371 km, Volume of Earth ~ $1.083 * 10^{21}$ m^3, Volume of Troposphere ~ $6.13 * 10^{18}$ m^3]
Which is the correct volume of CO_2 in the troposphere?
A. $2.453 * 10^{18}$ m^3
B. $2.453 * 10^{20}$ m^3
C. $2.453 * 10^{22}$ m^3
D. None of the above

Total mass of troposphere $\rho = \frac{m}{V}$

M_a (atmosphere) $\rho =$

$\frac{M_T}{V_T} = \frac{M_A}{V_A}$

$M_T = 0.8 M_A$

$V_A = \frac{M_A}{M_T} V_T = \frac{1}{0.8} V_T$

Total CO_2 in Earth atm $= \frac{0.04}{100} \times V_{atm} = \frac{1}{0.8} \times V_T \times \frac{0.04}{100}$

26. Biodegradation refers to:
A. Breakdown of organic matter by microorganisms
B. Conversion of organic matter into polymers
C. Chemical synthesis of pollutants
D. Physical removal of pollutants

27. Bioremediation primarily involves the use of:
A. Heavy metals to remove pollutants
B. Microorganisms and plants to degrade or detoxify pollutants
C. High-temperature combustion only
D. Organic solvents

28. Which microorganisms can convert pollutants into CO_2, H_2O, microbial biomass, or less-toxic compounds?
A. Only bacteria
B. Only fungi
C. Bacteria, fungi and algae
D. Only algae

29. Biodegradation of oil primarily occurs at the:
A. Air–soil interface
B. Oil–water interface
C. Rock–water interface
D. Soil–air interface

30. Which enzyme activates molecular oxygen during microbial degradation of hydrocarbons?
A. Lipase
B. Cytochrome P-450
C. Amylase
D. Protease

31. β-Oxidation involves stepwise oxidation of the fatty acyl group at the:
A. α-position
B. β-position
C. γ-position
D. Terminal methyl group

32. Oxidation of benzene with air over a catalyst at 2–5 bar and 400–450 °C produces:
A. Phthalic acid
B. Maleic anhydride
C. Catechol
D. Salicylic acid

33. Which compound is a byproduct in the chemical oxidation of benzene?
A. Fumaric acid
B. Glycine
C. Hippuric acid
D. Phthalic acid

34. Oxidation of naphthalene using O_2/V_2O_5 produces:
A. Phthalic acid
B. Phthalic anhydride
C. Phthalaldehyde
D. 1,4-Naphthoquinone

35. The first metabolite formed during methanol oxidation is:
A. Acetic acid
B. Formaldehyde
C. Acetaldehyde
D. Acetyl-CoA

36. Which of the below statements is/are correct regarding green chemistry?
A. It focuses on pollution prevention at the molecular level
B. It promotes hazardous substances
C. It aims to reduce or eliminate hazardous substances
D. Both A and C

37. Green chemistry primarily focuses on:
A. Treating pollution after it is produced
B. Preventing pollution at the molecular level
C. Increasing chemical waste
D. Increasing disposal

38. The first principle of green chemistry is:
A. Atom economy
B. Prevention
C. Catalysis
D. Design for degradation

39. The principle of atom economy aims to:
A. Maximize incorporation of reactants into the desired product
B. Maximize solvent consumption
C. Increase synthetic steps
D. Increase byproducts

40. Design for Energy Efficiency recommends:
A. Very high pressure
B. Maximizing heating and cooling
C. Minimizing energy requirements
D. Increasing distillation

41. Which is a renewable feedstock?
A. Coal
B. Petroleum
C. Biomass
D. Natural gas

42. Which principle recommends avoiding unnecessary protection/deprotection steps?
A. Reduce derivatives
B. Catalysis
C. Prevention
D. Atom economy

43. Catalytic reagents are preferred over stoichiometric reagents because they generally:
A. Increase waste
B. Improve efficiency and reduce material consumption
C. Always require higher temperature
D. Increase reaction steps

44. Design for Degradation requires chemical products to:

A. Persist indefinitely
B. Form toxic products
C. Break down into innocuous degradation products
D. Become more persistent

45. Inherently safer chemistry for accident prevention aims to minimize:
A. Product yield
B. Chemical accidents
C. Atom economy
D. Renewable feedstock usage

46. Atom economy is calculated as:
A. Mass of reactants / mass of product × 100
B. Mass of desired product / total mass of reactants × 100
C. Actual yield / theoretical yield
D. Waste / product × 100

47. Which reaction type can theoretically approach 100% atom economy?
A. Substitution
B. Elimination
C. Addition
D. Reaction producing large stoichiometric byproducts

48. The traditional industrial synthesis of ibuprofen was less green mainly because it:
A. Used fewer renewable feedstocks
B. Was a six-step process producing significant unwanted byproducts
C. Had less atom economy
D. Used solvents

49. The conventional production of ethylene oxide used:
A. Molecular oxygen only
B. A chlorohydrin intermediate
C. Biomass fermentation
D. Carbon dioxide

50. Industrial ecology aims primarily to:
A. Maximize non-renewable resource consumption
B. Minimize environmental impact while optimizing resource, energy and capital utilization
C. Eliminate industrial production
D. Increase waste disposal